# T14 Flat Price Estimator

Final Project Submission

**Objective:** Build an end-to-end machine-learning system that estimates residential flat prices in Rs lakh and deploy it through a Streamlit application.

## 1. Project Overview

The T14 Flat Price Estimator predicts the estimated price of a residential flat using property characteristics such as location, size, BHK, building age, floor, furnishing, parking, amenities, and distance to metro/highway. The final system contains a trained Random Forest regression pipeline saved as **flat_price_model.pkl** and a Streamlit user interface.

## 2. Dataset

The correct worksheet was **Listings** from **T14_Flat_price_estimator.xlsx**. The project used 600 listings. The ID column and target were excluded from the model feature matrix.

| Dataset item | Actual result |
|---|---|
| Worksheet | Listings |
| Rows | 600 |
| Target | price_inr_lakh |
| ID excluded | listing_id |
| Model input features | 11 |

## 3. Features Used

| Numerical features | Categorical features |
|---|---|
| bhk | city |
| carpet_area_sqft | locality |
| building_age_years | furnishing |
| floor | parking |
| total_floors | |
| amenities_count | |
| distance_to_metro_or_highway_km | |

## 4. Data Cleaning and Validation

The dataset was checked for missing values, duplicate rows, data types, categorical values, suspicious numerical values, non-positive BHK and carpet area, negative building age, invalid floor values, invalid total floors, negative amenities, and floor/total-floor consistency. The original dataset was not modified during these checks.

## 5. Exploratory Data Analysis

Prices ranged from **Rs 24.5 lakh to Rs 291.6 lakh**. The mean price was **Rs 89.91 lakh** and the median was **Rs 80.05 lakh**.

| City | Count | Mean price (Rs lakh) | Median (Rs lakh) | Min | Max |
|---|---|---|---|---|---|
| Pune | 199 | 116.81 | 108.50 | 30.30 | 291.60 |
| Nashik | 200 | 77.18 | 75.90 | 24.50 | 186.10 |
| Nagpur | 201 | 75.94 | 68.30 | 27.80 | 179.20 |

**Average price by BHK**

| BHK | Count | Mean price (Rs lakh) | Median (Rs lakh) |
|---|---|---|---|
| 1 | 119 | 48.24 | 45.10 |
| 2 | 278 | 81.97 | 75.80 |
| 3 | 161 | 119.42 | 107.80 |
| 4 | 42 | 147.39 | 139.10 |

## 6. Feature Matrix and Train-Test Split

The feature matrix contained 600 rows and 11 input features. The target contained 600 values. An 80/20 split with **random_state=42** produced 480 training records and 120 testing records.

| Item | Shape / value |
|---|---|
| X | (600, 11) |
| y | (600,) |
| X_train | (480, 11) |
| X_test | (120, 11) |
| y_train | (480,) |
| y_test | (120,) |

## 7. Preprocessing

Numerical features used **StandardScaler**. Categorical features used **OneHotEncoder(handle_unknown="ignore")**. Both were kept with the regression estimator in the fitted pipeline.

## 8. Regression Model

| Model setting | Actual value |
|---|---|
| Algorithm | RandomForestRegressor |
| n_estimators | 300 |
| random_state | 42 |

## 9. Model Evaluation

Evaluation was performed on the actual 120-record test set.

| Metric | Actual result |
|---|---|
| MAE | Rs 8.21 lakh |
| RMSE | Rs 10.62 lakh |
| $R^2$ | 0.9449 |

## 10. Feature Importance

These are the actual top 10 encoded feature-importance results from the trained Random Forest. These values represent model influence, not causation.

| Rank | Feature | Importance |
|---|---|---|
| 1 | num__carpet_area_sqft | 0.498999 |
| 2 | cat__city_Pune | 0.155530 |
| 3 | cat__locality_Kothrud | 0.076614 |
| 4 | cat__locality_Wagholi | 0.058442 |
| 5 | num__building_age_years | 0.026341 |
| 6 | cat__locality_Baner | 0.025661 |
| 7 | cat__locality_Dharampeth | 0.022728 |
| 8 | cat__locality_Hinjewadi | 0.016911 |
| 9 | cat__locality_Kharadi | 0.016078 |
| 10 | cat__locality_College Road | 0.012684 |

## 11. Saved Model Verification

The complete fitted preprocessing and regression pipeline was saved as **flat_price_model.pkl**. It was reloaded successfully and produced the same prediction as the original fitted pipeline for a verification sample.

## 12. Streamlit Application

The Streamlit application uses the saved model and the same 11 input features. It provides inputs, validates user values, displays the predicted price in Rs lakh, and includes a buyer-friendly “What matters most?” section based on the actual feature-importance results. The application does not retrain the model.

| Application component | Implementation |
|---|---|
| Model | flat_price_model.pkl |
| Input features | Same 11 features as trained model |
| Output | Predicted price in Rs lakh |
| Validation | Carpet area > 0; BHK > 0; floor <= total floors |
| Buyer insights | Top actual feature-importance results |

## 13. Testing and Final Validation

The completed system was tested using normal, minimum, maximum, categorical, unseen-category, missing-value, invalid-value, and saved-model scenarios.

| Test | Observed result |
|---|---|
| Normal input | Prediction generated: Rs 145.83 lakh |
| Minimum values | Prediction generated: Rs 73.12 lakh |
| Maximum values | Prediction generated: Rs 245.83 lakh |
| Unseen categories | PASS - model handled unseen categories; Rs 70.42 lakh |
| Missing value | Model accepted missing value; Rs 140.21 lakh |
| Invalid values | Model produced Rs 77.11 lakh; Streamlit validation should prevent these inputs |
| Saved model | PASS - flat_price_model.pkl loaded and generated predictions |
| Final validation | PASS - model, features, pipeline, app and consistency checks completed |

## 14. Limitations

The model is based on the available 600 listings and may not represent every property market. Predictions are estimates rather than exact market prices. Feature importance describes model behavior rather than causal relationships. Real-world property prices may depend on factors not included in the dataset, and performance may change on substantially different data.

## 15. Conclusion

The T14 Flat Price Estimator successfully combines data preprocessing, exploratory analysis, Random Forest regression, model evaluation, feature-importance analysis, model persistence, testing, and Streamlit deployment. The final model achieved **MAE of Rs 8.21 lakh, RMSE of Rs 10.62 lakh, and $R^2$ of 0.9449** on the test set. The deployed application successfully uses the saved pipeline to generate flat-price estimates while explaining the factors most influential to the model's predictions.

# T14 Flat Price Estimator — Rubric Alignment Addendum

## Business Insight & Actionable Recommendation

Buyer recommendation: When comparing flats, buyers should pay particular attention to carpet area because it has the highest feature importance in the model at approximately **0.499**, substantially higher than the next feature, Pune city at **0.156**. Locality indicators such as Kothrud (**0.077**) and Wagholi (**0.058**) also contribute meaningfully to model predictions. These values represent model influence, not causal effects.

## Responsible AI & Personal Reflection

The model should be used as a price-estimation aid rather than a definitive valuation tool. Feature importance represents model influence and does not establish causation. Predictions may be less reliable for property types or categories that are poorly represented in the training data.

Through this project, I learned that iterative prompting was useful for debugging, validation, UI improvement, testing and documentation. However, AI-generated outputs still needed to be checked against the actual dataset and model results. I used the actual calculated metrics, feature-importance values and validation outputs rather than accepting unsupported AI-generated numbers.

## Rubric Coverage Check

| Rubric area | Evidence in project |
|---|---|
| Working build | Working Streamlit estimator using the saved flat_price_model.pkl pipeline. |
| Correctness & verification | Actual MAE, RMSE and $R^2$; saved-model reload check; input testing; final validation. |
| Prompting & iteration | 15 documented AI-use prompts with changes, reasons and evidence screenshots. |
| Business insight | Quantified feature-importance explanation and buyer-facing recommendation. |
| Documentation & reflection | AI-use log, limitations, responsible-AI note and personal reflection. |